



Exercise Sheet 05

Published: June 11, 2025

Due: June 18, 2025

Total points: 12

Please upload your solutions to WueCampus as a scanned document (image format or pdf), a typesetted PDF document, and/or as a jupyter notebook.

1. From Propositions to Probabilities

(a) Determine whether the set $\mathcal{F}$ is a valid set of events for a probability space $(\Omega, \mathcal{F}, P)$ with $\Omega = \{1, 2, 3, 4, 5\}$. If it is not valid, complete $\mathcal{F}$ with a minimum number of missing events.

2P

(i) $\mathcal{F} = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, \Omega\}$

(ii) $\mathcal{F} = \{\emptyset, \Omega\}$

(iii) $\mathcal{F} = \{\emptyset, \{1, 4\}, \{1, 2\}, \{1, 2, 4\}, \{2, 3, 5\}, \{3, 4, 5\}, \Omega\}$

(b) An unbiased coin is tossed twice.

2P

a) List all the possible outcomes.

b) What is the probability of obtaining two heads?

c) What is the probability of obtaining a head and a tail in any order?

(c) Two dice are rolled:

A = 'sum of two dice equals 3'

B = 'sum of two dice equals 7'

C = 'at least one of the dice shows a 1'

2P

a) What is $P(A|C)$?

b) What is $P(B|C)$?

c) Are A and C independent? What about B and C ?

2. Statistical inference

(a) Suppose that X takes values between 0 and 1 and has probability density function $2x$. Compute $\text{Var}(X)$ and $\text{Var}(X^2)$.

2P

(b) In a 90-minute football-match the average number of goals is 3. Suppose that goals are scored randomly over the whole time and are independent of each other. Calculate the probability of 4 goals in 6 minutes.

2P

(c) Consider a coin tossing scenario where a coin is tossed n times until the first head appears. The probability p of head is unknown. Use maximum likelihood estimation to infer model parameter p .

2P